BISHESH KAFLE

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EDUCATION

Technische Universität Dortmund — Dortmund, Germany
Master of Science in Data Science — Oct 2025 - Present
Current CGPA: 1.3 (51/120 credits completed)

Kathmandu University — Dhulikhel, Nepal
Bachelor of Science in Computational Mathematics — Aug 2019 - Aug 2024
Cumulative GPA: 3.93/4 (1.1 in German Scale)

WORK EXPERIENCE

Metro Markets GmbH — Düsseldorf, Germany
Data Scientist (Working Student) — Apr 2026 - Present

- Independently designed a model evaluation framework and implemented baseline models and selection strategies, reducing forecasting cost per run from €90 to €12 (87%) and runtime from approximately 7 to 2.5 hours (64%).
- Benchmarked TabICL and Amazon Chronos for demand forecasting, evaluating model performance and suitability for integration into the forecasting pipeline.
- Implemented and tested demand-forecasting pipeline components in Python, using ClickHouse, Dagster, and Docker to support model evaluation and workflow orchestration.
- Documented forecasting methodologies and presented model evaluation results to communicate performance findings and support technical decision-making.

Extensodata Pvt. Ltd. — Lalitpur, Nepal
Associate Data Analyst — Aug 2024 - Sep 2025

- Applied clustering, regression analysis, and hypothesis testing to customer segmentation in a fintech environment, supporting analysis for targeted marketing.
- Designed and optimized SQL-based data structures to support financial analytics.
- Developed Python ETL pipelines and data-quality checks for analytical datasets.

Data Intern — Apr 2024 - July 2024

- Explored customer segmentation, fraud detection, and Bayesian networks through data science research and development tasks.
- Developed models for regression, time-series analysis, and collaborative filtering.

RESEARCH AND PROJECTS

Evaluating XGBoost Algorithm for Nepali Stock Price Prediction (Conference Paper: ICT-CEEL 2023)

- Engineered technical indicators, including SMA (5), RSI, and MACD, and analyzed their importance in XGBoost-based stock-price prediction.
- Developed a predictive model leveraging the XGBoost algorithm to forecast stock prices using a time series dataset.

A Comparative Study of Principal Component Analysis' Impact on Performance Metrics of Regression Models

- Evaluated PCA-based dimensionality reduction across Linear Regression, Logistic Regression, and XGBRegressor using a 10,000-row and 201-column dataset, examining trade-offs between explainable variability and computational efficiency.

Case Study on Sexual Assault and Abuse in Nepal

- Analyzed data from NGOs and Nepal Police to visualize the geographic and temporal distribution of reported sexual assault and abuse cases in Nepal.
- Developed data-driven insights to communicate observed patterns and trends to aid policymakers, law enforcement, and advocacy groups in resource allocation and preventive planning.

SKILLS AND CERTIFICATIONS

Programming Languages: Python, SQL, R, MATLAB
Machine Learning & Statistics: Statistical modelling, Time-series econometrics & forecasting, Feature engineering, Model evaluation, Model selection
Tools: ClickHouse, Dagster, Apache Airflow, Docker, Git.
Visualization & Documentation: Matplotlib, Seaborn, Tableau, Streamlit, Markdown, LaTeX
Soft Skills: Communication, Teamwork, Critical Thinking, Adaptability, Time Management
Certifications & Training: Deep Learning using Supercomputer; Bayesian Statistics: From Concepts to Data Analysis; Advanced Learning Algorithms; Data Science Methodology; MATLAB Onramp.
Languages: German (A2), English (C1), Nepali (Native), Hindi (Native).

ACTIVITIES

Junior Optimist International Club / Young Adult Optimist International Club — Jun 2018 - Aug 2025
Co-Founder

- Also President, Junior Optimist International Club of Pokhara (Aug 2018–Aug 2019); Vice-President, Young Adult Optimist International Club (Apr 2024–Aug 2025).
- Conducted online talk shows on various global topics, such as climate change, mental health awareness, and youth empowerment, bringing together experts and young leaders to share insights and drive meaningful discussions.

Dare to Revolution Foundation — Jun 2018 - Dec 2019
Co-Founder

- Empower youth through workshops and training on public speaking, leadership and entrepreneurship.
- Promote local culture through a variety of programs and initiatives, aiming to preserve and celebrate traditional practices, art, and heritage.